

MATHEMATICS 2210
Quiz 7, Fall semester 2012

Name:.....

1. (6 pts) Find maxima and minima of $f(x, y, z) = 2x - 3y + 4z$ subject to the constraint $x^2 + y^2 + z^2 = 1$.

$$\nabla f = (2, -3, 4) \quad \nabla g = (2x, 2y, 2z)$$

$$\begin{cases} 2 = \lambda 2x \\ -3 = \lambda 2y \\ 4 = \lambda 2z \\ x^2 + y^2 + z^2 = 1 \end{cases} \quad x = \frac{1}{\lambda} = -\frac{2}{3}y = \frac{z}{2} \Rightarrow y = -\frac{3}{2}x \text{ \& } z = 2x$$

$$\Rightarrow x^2 + \frac{9}{4}x^2 + 4x^2 = 1 \Rightarrow \frac{29}{4}x^2 = 1 \Rightarrow x = \pm \sqrt{\frac{4}{29}} = \pm \frac{2}{\sqrt{29}}$$

$$y = \mp \frac{3}{\sqrt{29}} \quad z = \pm \frac{4}{\sqrt{29}} \quad \pm \frac{1}{\sqrt{29}}(2, -3, 4)$$

$$+ \left(\frac{2}{\sqrt{29}}, -\frac{3}{\sqrt{29}}, \frac{4}{\sqrt{29}} \right) = \frac{29}{\sqrt{29}} = \sqrt{29} \quad \underline{\text{MAX}}$$

$$+ \left(-\frac{2}{\sqrt{29}}, \frac{3}{\sqrt{29}}, -\frac{4}{\sqrt{29}} \right) = -\sqrt{29} \quad \underline{\text{MIN}}$$

2. (5 pts) Find the volume of the solid under $z = x \cos(xy) + 5$ over the region $R = [0, \pi/12] \times [0, 3]$.

$$\int_0^{\pi/12} \int_0^3 (x \cos(xy) + 5) dy dx = \int_0^{\pi/12} [\sin xy + 5y]_0^3 dx = \int_0^{\pi/12} \sin 3x + 15 dx$$

$$= \left[-\frac{1}{3} \cos 3x + 15x \right]_0^{\pi/12} = -\frac{1}{3} \cos \pi/4 + \frac{15\pi}{12} + \frac{1}{3} \cos 0 = -\frac{1}{3} \frac{\sqrt{2}}{2} + \frac{5\pi}{4} + \frac{1}{3}$$